

Full length article

Enhancement of ultrashort laser pulses absorption in glass using single MHz burst

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ABSTRACT

In this study, the effects of burst mode on the absorption of ultrashort laser pulses in glass are investigated. The material modifications induced by single pulses and by single bursts with the same energy are observed in optical transmission microscopy. 40 μ J bursts with different numbers of pulses are considered. Our results demonstrate that distributing energy over multiple pulses in a burst (with an intra-burst frequency of 40 MHz) significantly enhances the material modification. Optical microscopy confirms that burst mode modifications produce material alterations whereas a single-pulse irradiation does not. There is an optimal pulse number after what the modifications decrease. We show how a glass-to-glass welding can be achieved in burst mode with an energy for which it is not possible using a single pulse. These findings highlight the potential of burst mode to improve laser processing of transparent materials using less power.

1. Introduction

The development of ultrashort pulse lasers has led to numerous innovations and industrial applications [1,2]. Among these, the micro-manufacturing of transparent materials has greatly benefited from the high peak powers delivered by such lasers. Specifically, the peak intensity of ultrashort pulses enables non-linear interactions, such as multiphoton absorption, within the material volume [3,4]. As a result, it becomes possible to fabricate waveguides within the bulk of glass [5,6], store data in three, four and even five dimensions [7–9], create micro-channels [10] and even weld glass for applications across electronics, photonics, microfluidics, and more [11–13]. Furthermore, processes based on ultrashort non-linear absorption are typically highly localized, offering high-precision machining without damaging surrounding areas.

Recently, burst mode has emerged as a novel method for temporally distributing laser pulses, opening up new possibilities. In burst mode, laser energy is delivered as a finite sequence of closely spaced pulses, referred to as a burst, with intra-burst repetition rates ranging from several MHz up to GHz, while the inter-burst repetition rate remains comparatively lower. This technique has already demonstrated

advantages in various applications, including the enhanced efficiency of metal ablation [14,15].

In this paper, we demonstrate that burst mode is also advantageous for efficient energy deposition in transparent materials. Our results show that distributing the same amount of energy over a burst of pulses leads to greater energy deposition in glass than using a single pulse. While previous studies have compared modifications in glass induced by single pulses versus MHz or GHz bursts [16], they did not distinguish between intra-burst effects and inter-burst cumulative effects due to high repetition rates yielding to burst overlap. In our approach, we employ no overlap between bursts, allowing each burst's effect to be clearly isolated. By limiting the total energy deposited at each point, we avoid the formation of cracks reported in other studies [16,17]. This suggests that burst mode could enable similar or improved processing outcomes while requiring less laser power compared to single-pulse operation, a conclusion supported by the results of K. Mishchik *et al.* for glass cutting [18] and we here demonstrate for glass welding.

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2. Experimental setup

To assess whether burst mode enhances the material modifications, we compare the effects of a single pulse with those of a burst containing the same total energy. If the glass exhibits more pronounced modifications when the energy is delivered in burst form, this shows that the burst mode enables more efficient energy deposition.

The experiments were conducted using an ultrafast laser operating at a wavelength of 1030 nm (Satsuma X, Amplitude) with a pulse duration fixed at 1.5 ps (measured by an autocorrelator PulseCheck NX 50). The laser beam was focused into the volume of a $50 \times 20 \times 5$ (length $\times$ height $\times$ width) mm³ Borofloat® 33 glass from Schott, 3 mm below the top surface (taking the optical refractive index into account), using a 50 mm F-theta lens. The data concerning the glass composition and its physical properties are available in Table 1. The band gap energy has been calculated using the Tauc's plot technique [19,20] with the transmittance data furnished by Schott. The band gap energy has been found to be equal 4.97 eV. As the 1030 nm photons have an energy equal to 1.20 eV, it means that 5 photons are absorbed simultaneously during the multi-photoionization process.

The optical path of the laser beam is illustrated in Fig. 1. The scanning speed was set high enough to prevent an overlap between successive pulses or bursts. Given a measured spot diameter of 30 μ m in the air and a repetition rate of 10 kHz, a scanning speed of 300 mm/s ensures no overlap (or 0 % recovery rate). This configuration guarantees that the observed modifications arise solely from the effects of individual pulses or bursts because each of them will be separated by a large distance within the material.

The Fig. 2 shows how this no overlap is used to create modifications within the glass using only a single pulse or only a single burst. It has to be noted that because of the very low time delay between the pulses of the same burst compared to the delay between the bursts, the pulses within the same burst virtually land on the same spot (the 25 ns delay between each pulse of the burst causes a shift of only 7.5 nm between two pulses of the same burst when a speed of 300 mm/s is used).

In the experiment, a first linear scan was achieved using a discrete series of single-pulse spots at 40 μ J per pulse. This was followed by a second linear scan composed of burst spots, each consisting of two pulses with a combined energy of 40 μ J. This pattern was continued such that the Nth line was achieved using a burst of N pulses, all with the same total energy of 40 μ J per burst. In total, 30 linear scans were performed, ranging from single pulses up to 30 pulses per burst (ppb), with an intra-burst repetition rate of 40 MHz. Each line was separated by 50 μ m. The experimental layout is summarized in Fig. 3. Each line has been carried out with an initial temperature equal to the room temperature. The total energy of 40 μ J has been chosen based on a threshold test: at this energy level, no visible modification was observed with a single pulse, while a 45 μ J pulse produced a detectable mark. This threshold ensures that any observed modifications in the burst regime result from improved energy deposition efficiency.

The energy distribution within each burst was measured using a Thorlabs DET10A2 photodiode. Fig. 4 displays the power distribution

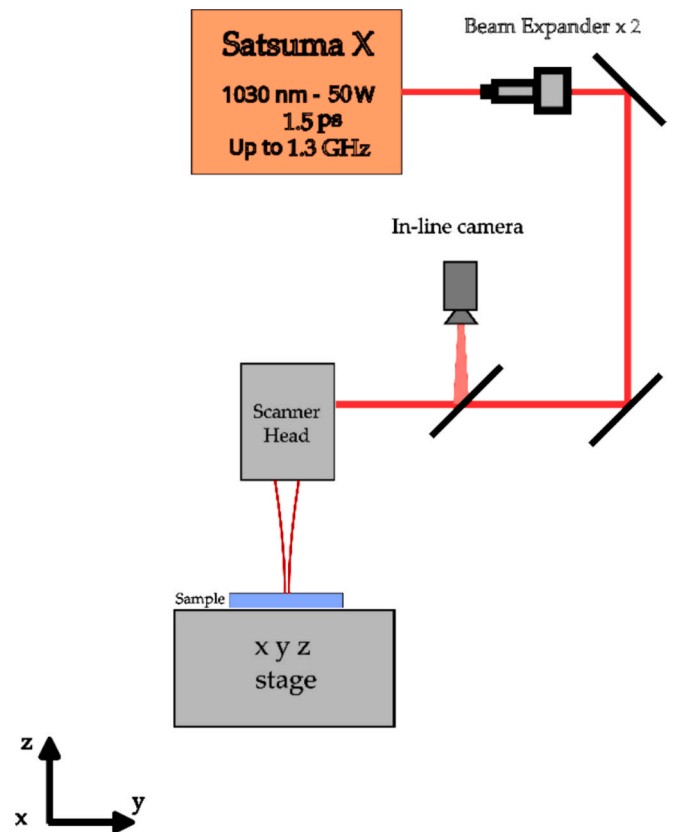


Fig. 1. Experimental setup for the glass modification and the glass welding.

for different bursts. The first pulse carries the highest energy due to gain depletion, with subsequent pulses decreasing progressively in intensity.

3. Results and discussion

After laser irradiation, the sample was examined using an optical transmission microscope, both in standard transmission mode and between a polarizer and an analyzer (Fig. 5). The microscope enabled the observation of the glass modification by the laser interaction. Additionally, imaging between crossed polarizers, a classical technique for enhancing contrast, was used to reveal birefringence patterns. These anisotropies arise from localized stress fields generated within the glass volume by the laser-induced modifications [21].

A clearly visible continuous line at both the top and bottom of the sample marks the boundaries of the test area. Within this region, point-like modifications can be observed in the glass volume. These modifications do not all lie on the same horizontal plane. This variation in depth is due to the optical Kerr effect, which causes self-focusing of the more energetic pulses, bringing the focal point closer to the geometrical focus. As the number of pulses in the burst increases, the energy of individual pulses decreases progressively, decreasing the optical Kerr effect, resulting in shift of the focal points deeper into the glass.

To capture the full range of these modifications, the images in Fig. 5 were taken at different depths using the microscope. The early lines (with fewer pulses per burst) appear closer to the upper surface, while the later lines (with more pulses per burst) are located deeper, consistent with the influence of the Kerr effect. Moreover, adding pulses within a burst completely changes the absorption geometry and the plasma dynamic within the glass as the plasma created by the first pulse will provoke scattering of the incident light of the next pulses, thus modifying the focus position even more, as discussed by Bulgakova and al [22,23].

The first and second series at the top of the sample are nearly

Table 1

Chemical composition and physical properties of the Borofloat® 33 glass from Schott.

Composition of the glass					
Element	SiO ₂	B ₂ O ₃	Na ₂ O	Al ₂ O ₃	K ₂ O
Proportion [%]	81	13	3	2	1
Physical properties					
Optical index at 1030 nm	1.462				
Direct Band Gap energy	4.97 eV				
Density	2.23 g/cm ³				
Glass transition temperature	525 °C				
Young modulus	64 GPa				

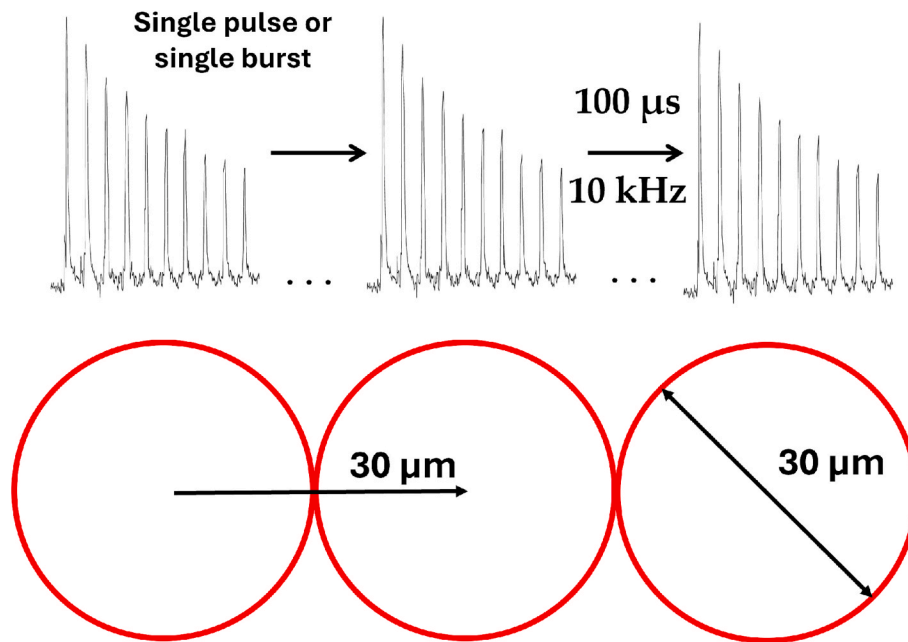


Fig. 2. Principle of the use of the high speed (300 mm/s) and low repetition rate (10 kHz) in order to have single pulse or single burst modification with no overlap.

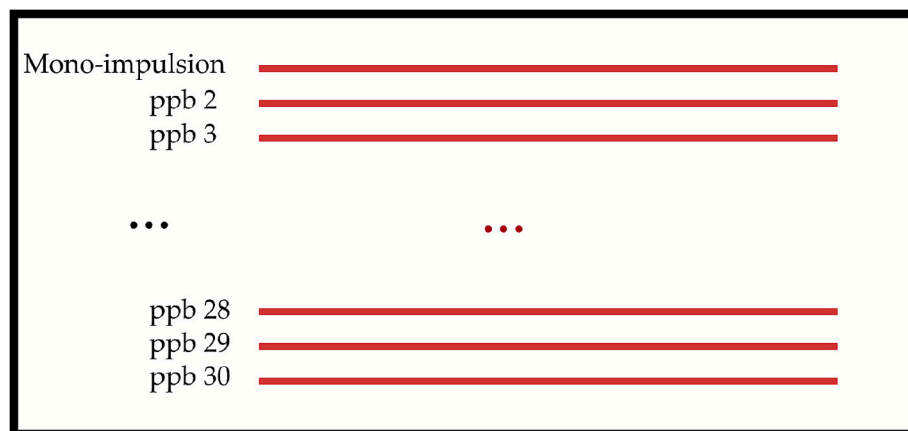


Fig. 3. Experiment layout on the glass plate (ppb: pulse per burst).

invisible, indicating that the laser pulses at these settings do not induce sufficient modification to be clearly detected under the optical microscope. However, when the same total energy per burst (40 μJ) is distributed across an increasing number of pulses, the resulting laser marks become progressively more visible. Modifications become distinctly contrasted at 4 pulses per burst (ppb) when viewed between two crossed polarizers, and at 5 ppb in standard transmission mode. As the number of pulses per burst continues to increase, the visibility of the modifications initially improves, reaching a maximum at around 15 ppb. At higher ppb values, the modifications begin to disappear but remain detectable up to 25 ppb. These observations demonstrate that distributing 40 μJ of energy across 4 to 25 pulses (at a 40 MHz intra-burst frequency) enhances the effectiveness of the energy deposition, resulting in more pronounced material changes compared to single-pulse irradiation. Notably, single-pulse modifications are not visible under crossed polarizers but become clearly observable once the burst contains more than 3 pulses. It is true even though in burst mode a lower energy per pulse may induce in an unmodified material a lower non-linear absorption.

In the image of the sample seen between the crossed polarizer, it can be observed that the stress is located around the modification spot, and

that the distribution is clover leaf shaped. This effect is well documented in literature for the stress repartition around holes [24,25] and around densification of glass during waveguide writing by femtosecond laser irradiation [26–28]. It is the signature of radial stress, which induces optical anisotropy by elasto-optic effects. The linear polarizers break the symmetry.

The material modification seems to be optimal for the burst ranging from 10 to 20 ppb. The Fig. 6 shows the modifications from 14 to 17 ppb.

It can be seen that the spot seems to vary in size depending on the number of pulses per burst. This is due to the difference in position of the modification within the material. When the modifications are not in focus of the optical microscope, they appear larger and the black of the spots are caused by light scattering. It can also be seen that the size of the modifications is much smaller than the spot size, because only the part or the gaussian intensity above the absorption threshold can trigger modification. Only the tip of the gaussian distribution of the intensity can modify the material effectively. Moreover, the self-focusing of the pulses provokes a spatial tightening of the intensity distribution. Beyond 25 ppb, the modifications vanish, which can be attributed to the reduced energy per pulse—falling below the threshold required to trigger significant non-linear absorption effects. This confirms that using burst

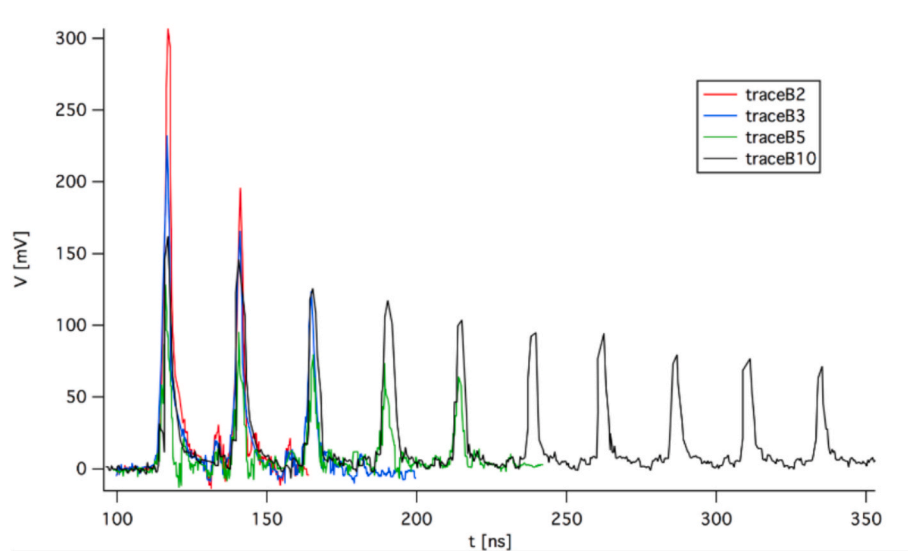


Fig. 4. Power distribution across a burst with 10 pulses measured with a photodiode at the exit of the laser system. Intra-burst frequency: 40 MHz.

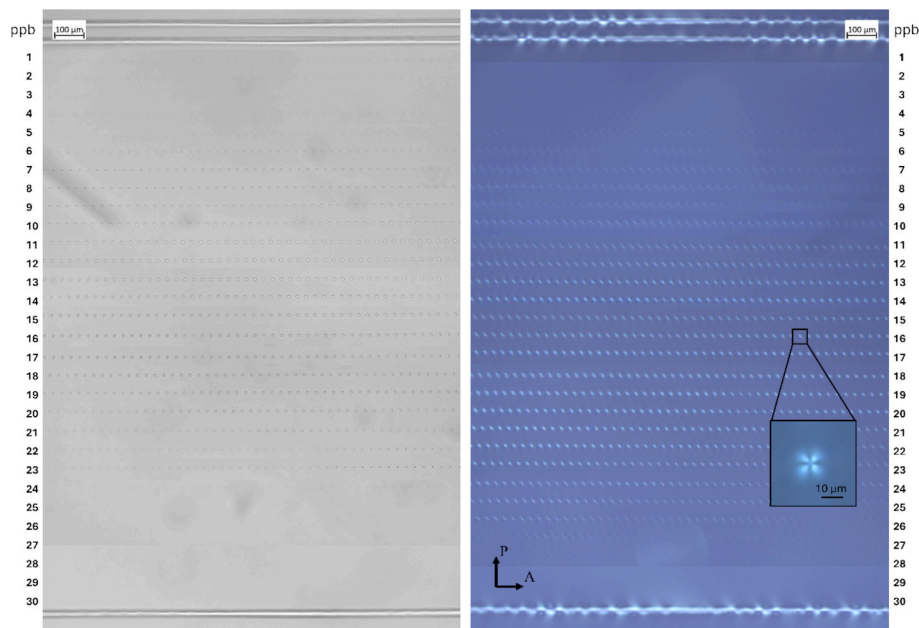


Fig. 5. Microscope transmission view of the modifications achieved in bulk borosilicate glass ($\times 5$ magnification). Left: simple view in transmission. Right: View between crossed polarizers. P: Polarizer direction. A: Analyzer direction. White light illumination. Single 40 μJ burst within each point. Only the number of pulses per burst changes. Intra-burst frequency: 40 MHz.

mode at 40 MHz enables more efficient energy deposition into the glass, maximizing structural modification for a fixed total energy. The most realistic explanation is that the initial pulses within a burst contribute to enhancing the absorption of the subsequent ones. It is known that thermal effects appear in the order of one nanosecond for dielectric materials [29]. So during the 25 ns between two pulses in the burst the lattice temperature has the time to rise, which increases the absorptivity of the material as reported in [18]. Another possible physical mechanism involves the generation and accumulation of free carriers induced by the first pulses that will act as incubators for subsequent pulses [30]. Even though the thermalization of the free carriers occurs generally in less than a nanosecond, the free carriers can live up to several microseconds in large band-gap materials such as SiO_2 [31]. Along with the temperature augmentation, they may also contribute to increase the absorptivity of the material during the remainder of the burst [9,23]. In

addition, as the number of pulses within the burst augments, the energy per pulse decreases, mitigating the optical Kerr effect. As the optical Kerr effect decreases, the non-linear intensity should also decrease and the modifications should be lesser. However, in spite of this, the modifications are greater when the energy is distributed across a burst.

To further investigate the absorption phenomenon, a measurement of the light intensity emitted by the material has been carried out. The sensor used is an Optronis SC-10 Systems. The Fig. 7 illustrates the result of the measurement for a 40 μJ burst with 10 ppb. The bold curve indicates that the material light emission is starting at the 6th pulse of the burst, even if the first pulses are the most energetic. And then the light intensity emitted by the material gets higher and higher even if the pulses are less and less energetic. The rapid decay of the light intensity (decay time of around 15 ns) indicates that the light emission cannot be solely attributed to glass temperature, as the characteristic thermal

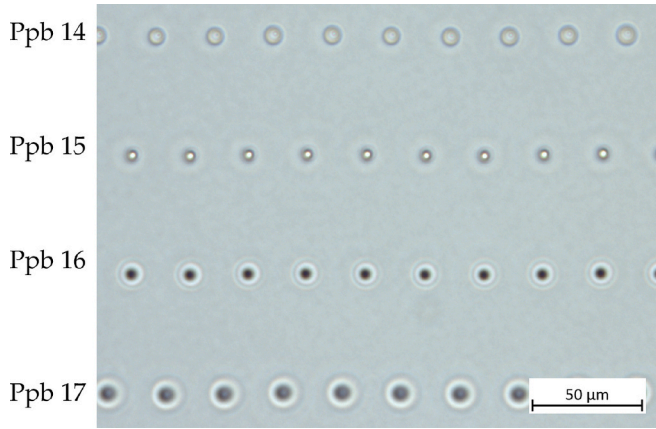


Fig. 6. $\times 20$ magnification of the modifications for the bursts with 14 to 17 ppb. The difference in appearance is due to a different depth position of the modifications. The 15 ppb line is the most in focus in this picture.

diffusion time has been calculated to be on the order of $3.5 \mu\text{s}$ in borosilicate. It is coherent with the hypothesis of free carriers accumulation and also with the creation of a plasma, which also could explain the blue light emission observed during the process. The material is in an excited state that causes it to emit light.

The light emission increase for each pulse after the 5th one (red arrows in figure 7) and their characteristic decay time has been measured for each exponential decay and is indicated in Table 2. Contrary to the classical thermal accumulation, the augmentation of the light intensity is increasing for each pulse. Indeed, for conventional thermal accumulation [11], intensity increases should be smaller and smaller from pulse to pulse. On the contrary, Fig. 7 shows larger and larger increases. For example, pulse $n^\circ 7$ increases emissions by 0.32 ua, whereas pulse $n^\circ 6$, despite being more energetic, only increases them by 0.15 ua.

The modifications have also been observed from the side from the single pulse to the 30 pulses per burst, results are displayed in the Fig. 8. The laser beam propagates from the left side of the figure to the right side. The direction is indicated with the wave vector $\mathbf{k}$. No modifications in the material are visible in the first few lines, whereas the trace of the burst with 5 pulses can be seen. First the modifications extend, to reach a

maximum length of about $500 \mu\text{m}$ for the bursts of 10, 11 and 12 pulses per burst, and then the length of the modifications are shortening again to completely disappear at 28 pulses per burst. The expected focus shift is clearly visible. The more pulses in the burst, the less self-focusing is happening because the energy of each pulse is lowered. With the $\times 20$ magnification of the region of interest (ROI), it is possible to estimate that the diameter of the modification due to the burst of 16 pulses is about $6 \mu\text{m}$. There are no voids nor cracks, and the teardrop effect is also not observed. However, a clear double focalization effect is observed. This is attributed to filamentation effect, and this phenomenon is extensively documented in literature [32–34]. This double structure is produced because the self-focusing by optical Kerr effect happens and a first focalization enables non-linear absorption, leading to material modification. When the intensity is high enough to create a dense enough plasma, the beam is defocalized, and if the beam still has enough power, a second self-focalization occurs a new zone of modification appears.

The displacement of the modification relative to one another has been plotted in function of the number of pulses within the burst in Fig. 9, with the last burst visible as reference. It is in accordance with the Marburger theory [35].

To illustrate a practical application of such a burst mode, a glass-to-glass welding experiment was conducted using the exact same setup as described in Fig. 1. Two glass plates of Borofloat 33 of dimensions $50 \times 20 \times 5 \text{ mm}^3$ were set in optical contact prior to the welding, leading to a 10 mm thick stack (see Fig. 10).

The laser was focalized at the interface between the two glass plates. The welding parameters were similar to those used in the burst mode absorption tests. The scanner head operated at a speed of 100 mm/s (corresponding to $10 \mu\text{m}$ between each spot), with an inter-burst repetition rate of 10 kHz , a burst energy of $40 \mu\text{J}$, and an intra-burst frequency of 40 MHz . This setup corresponds to a mean power of 0.4 W .

Table 2

Value of the light intensity emitted by the material increase after the pulses of a 10 ppb, $40 \mu\text{J}$, 40 MHz burst.

Pulse number	6	7	8	9	10
Increase in light intensity (u.a)	0.15	0.32	0.40	0.46	0.57
Characteristic decay time (ns)	5.8	8.3	9.6	11,6	14

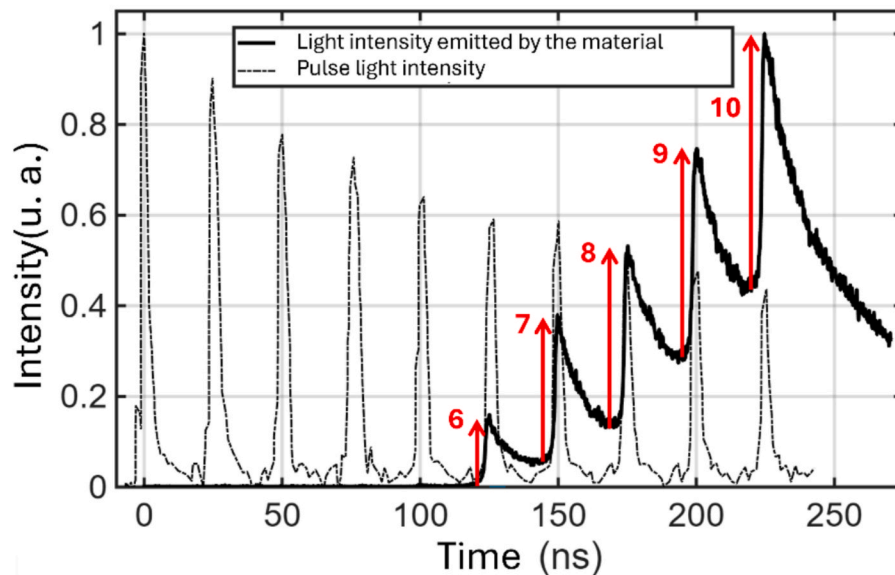


Fig. 7. Ultrafast measurement of the light intensity emitted by the borosilicate under a $40 \mu\text{J}$ burst with 10 ppb. Red arrows correspond to the increase of light intensity reported in Table 2.

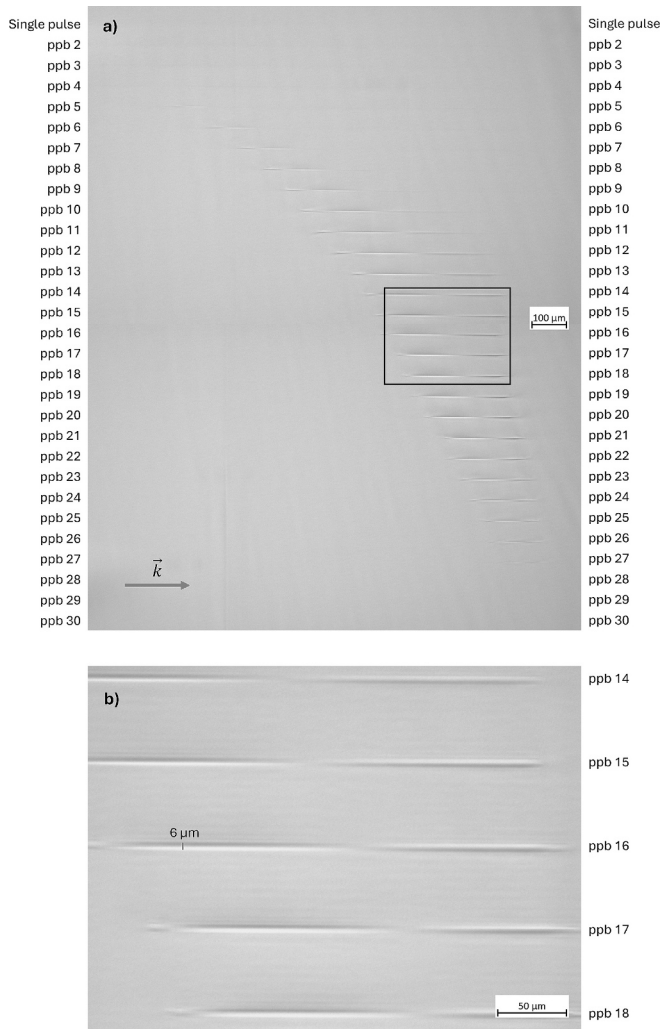


Fig. 8. Side view. Transmission microscopy image of the cross-section of the test, from single pulse to 30 pulses per burst. a) Magnification $\times 5$. b) Magnification $\times 20$ on the ROI.

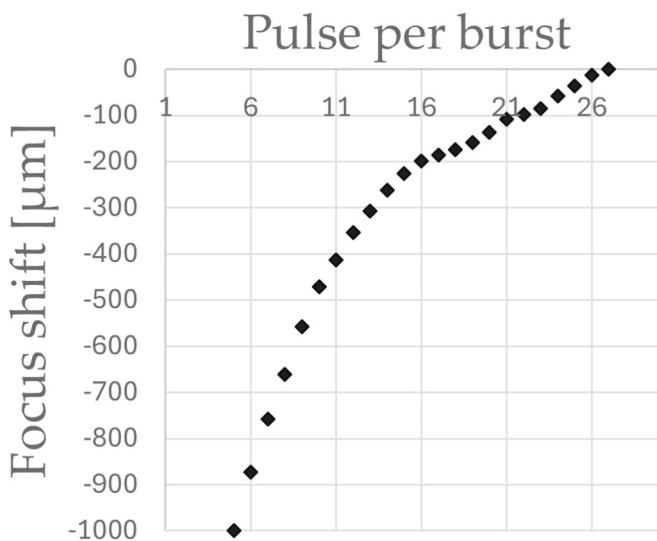


Fig. 9. Relative focus shift (optical Kerr effect) measured in function of the number of pulses within the burst of 40 μJ . The 0 is close to the ray tracing focusing point.

The number of pulses per burst was set to 15, as this value is located in the optimum zone of modification which extends approximatively between 10 to 20 ppb. The welding was achieved on a square area of $3 \times 3 \text{ mm}^2$ with a 20 μm hatch. Under these conditions, the two borosilicate glass plates were successfully welded together, as it is possible to see on the Fig. 11 thanks to the decontacted zone around the welded square.

The result was observed in transmission mode using a microscope at $\times 10$ magnification, as shown in Fig. 12.

Each visible point corresponds to a weld spot, with each spot created using a single burst. For comparison, the same welding attempt has been performed using single pulses with the same energy, but no successful bonding occurred. A mechanical resistance test has been carried out using a tensile stress machine (Shimadzu AGS-X). The breaking strength of the welding was 44.5 N. As the number of points have been calculated to be 45 000 (150×300) and the surface of a single point is $28.3 \mu\text{m}^2$, the total welded area was $1,27 \text{ mm}^2$ and therefore the tensile stress calculated is equal to 35 MPa which is comparable to literature [11,33]. The discrete nature of the welding has the advantage of preventing the crack propagation that can occur with continuous welding [36].

These results further support the conclusion that burst mode at 40 MHz significantly enhances the modifications in glass compared to single pulse. It highlights the practical advantage of burst mode in laser processing applications, such as precision welding of transparent materials.

4. Conclusion

We have demonstrated the effectiveness of burst mode for enhancing the material modification caused by the laser energy in glass. By distributing the same total energy across multiple pulses within a burst, significant improvements in material modifications were observed compared to single-pulse irradiation. With an intra frequency of 40 MHz, we showed that it is possible to optimize the number of pulses within the burst to improve the energy deposition, resulting in deeper and more pronounced modifications within the glass. For 40 μJ and 40 MHz bursts, the optimal results occurred between 10 and 20 pulses per burst. A study varying the intra-burst frequency should be carried out to find an optimal frequency.

A practical application of burst mode was demonstrated in a glass-to-glass welding experiment, where the laser's ability to enhance the modifications allowed for successful bonding of two glass plates. We couldn't achieve the same result with a single-pulse irradiation at the same energy level.

Overall, our findings suggest that burst mode offers a powerful tool for improving the efficiency and effectiveness of laser processing of transparent materials, reducing the overall energy requirements for achieving similar or enhanced outcomes.

CRedit authorship contribution statement

Baptiste De Azevedo: Writing – review & editing, Writing – original draft, Resources, Methodology, Investigation, Data curation, Conceptualization. **Jesse Schiffler:** Resources. **Thierry Engel:** Supervision, Data curation. **Frédéric Antoni:** Supervision. **Wilfried Uhring:** . **Éric Audouard:** Supervision, Project administration, Methodology, Conceptualization. **Armel Bahouka:** Writing – original draft, Supervision, Project administration, Conceptualization. **Sylvain Lecler:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Sylvain Lecler reports financial support was provided by French

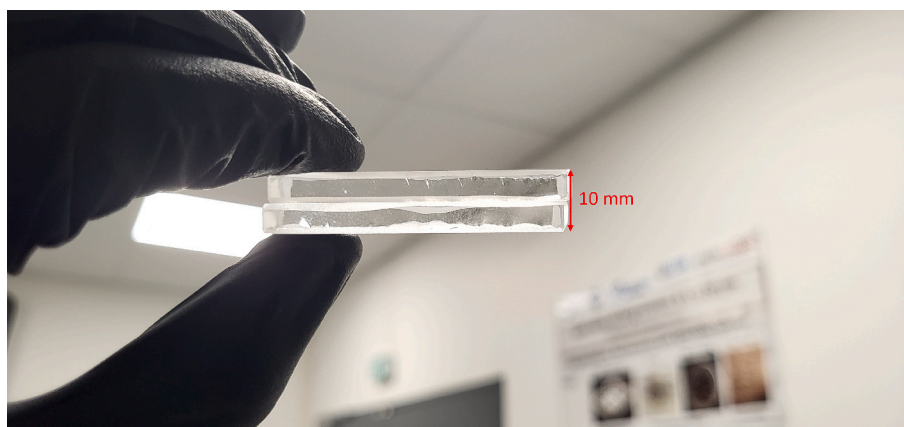


Fig. 10. Borosilicate glass stack to be welded.

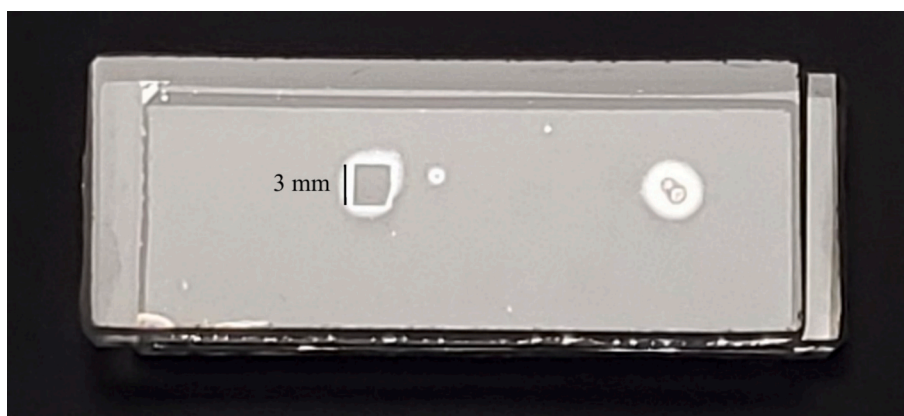


Fig. 11. Top view of the glass stack after the welding. The welded square is visible thanks to the decontacted zone around it.



Fig. 12. View of the welding spots between two borosilicate glass plates. Welding parameters: 100 mm/s, 10 kHz, 15 ppb, 40 μ J/burst, 40 MHz intra-burst frequency, 0.4 W, hatch: 20 μ m. $\times 10$ magnification.

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Data availability

Data will be made available on request.

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